

Retail Store Sales Analysis Live — Auto Analysis

Generated on 2026-09-24 11:45 UTC

Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Retail Store Sales Analysis Live".

Dataset Overview

File	retail_sales.csv
Rows	500
Columns	14
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	gradient_boosting
Problem Type	regression
Target Column	sales_amount

Metrics

Metric	Value
mae	27.3939
mse	1512.5076
rmse	38.891
r2	0.991
mape	0.1159

actual_vs_predicted	<pre>[{'actual': 1066.8, 'predicted': 1112.0213}, {'actual': 851.19, 'predicted': 779.7062}, {'actual': 1120.2, 'predicted': 1083.8651}, {'actual': 1427.48, 'predicted': 1349.9786}, {'actual': 757.61, 'predicted': 696.6401}, {'actual': 456.32, 'predicted': 481.4373}, {'actual': 185.06, 'predicted': 196.7306}, {'actual': 687.33, 'predicted': 654.5492}, {'actual': 243.66, 'predicted': 222.0729}, {'actual': 473.18, 'predicted': 485.6766}, {'actual': 424.27, 'predicted': 451.0532}, {'actual': 17.15, 'predicted': 11.4298}, {'actual': 445.3, 'predicted': 458.3255}, {'actual': 1309.68, 'predicted': 1191.1535}, {'actual': 145.81, 'predicted': 231.8855}, {'actual': 78.04, 'predicted': 83.971}, {'actual': 681.39, 'predicted': 690.2677}, {'actual': 1003.99, 'predicted': 1031.2429}, {'actual': 78.71, 'predicted': 98.6406}, {'actual': 288.08, 'predicted': 300.1074}, {'actual': 105.74, 'predicted': 110.1748}, {'actual': 110.41, 'predicted': 120.0049}, {'actual': 825.92, 'predicted': 804.9374}, {'actual': 223.51, 'predicted': 230.3834}, {'actual': 951.51, 'predicted': 851.1351}, {'actual': 660.43, 'predicted': 652.1997}, {'actual': 311.85, 'predicted': 315.9279}, {'actual': 980.72, 'predicted': 1031.8326}, {'actual': 40.19, 'predicted': 112.7518}, {'actual': 286.15, 'predicted': 264.511}]</pre>
residuals_sample	<pre>[-45.2213, 71.4838, 36.3349, 77.5014, 60.9699, -25.1173, -11.6706, 32.7808, 21.5871, -12.4966, -26.7832, 5.7202, -13.0255, 118.5265, -86.0755, -5.931, -8.8777, -27.2529, -19.9306, -12.0274, -4.4348, -9.5949, 20.9826, -6.8734, 100.3749, 8.2303, -4.0779, -51.1126, -72.5618, 21.639]</pre>
cv_scores	<pre>[0.9960953871235533, 0.9870467806452239, 0.9850597955222434, 0.9905615027174381, 0.9844053875609163]</pre>
cv_mean	0.9886
baseline	<pre>{'baseline_score': -0.0032, 'baseline_metric': 'r2', 'model_score': 0.991, 'improvement': 0.9942, 'cv_std': 0.0043, 'clearly_beats_baseline': True}</pre>
leakage_warnings	<pre>[{'column': 'profit', 'correlation': 1.0, 'reason': "Nearly identical to the target (100% correlated) — likely a direct calculation or duplicate of 'sales_amount', which would let the model 'cheat' by reconstructing the target instead of learning real patterns."}]</pre>

Feature Importance

Feature	Importance
unit_price	0.4904
quantity	0.4793
discount	0.0296
order_date_dayofweek	0.0002

store_city_Faisalabad	0.0
order_date_month	0.0
customer_rating	0.0
sales_channel_In-store	0.0
product_category_Groceries	0.0
product_category_Electronics	0.0
product_category_Clothing	0.0
returns	0.0
promotion	0.0
store_city_Lahore	0.0
product_category_Furniture	0.0
store_city_Multan	0.0
product_category_Toys	0.0
store_city_Islamabad	0.0
sales_channel_Online	0.0
store_city_Karachi	0.0

AI Business Insight AI-generated insight

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.